ALETHEIA: Technical White Paper

Exposing Funding Conflicts and Health Misinformation Through AI-Powered Analysis

Executive Summary

Aletheia is a browser extension and AI-powered research platform designed to **expose** funding conflicts, propaganda, and misinformation in health and medical content**.** In an era where powerful entities—pharmaceutical companies, food corporations, and special interests—shape public narratives for profit, Aletheia serves as a critical tool for independent research and truth-seeking.

The platform analyzes health articles when users manually click the extension icon, revealing:
• Who funds the publication (pharmaceutical companies, food industry, advertising)
• Conflicts of interest (corporate ownership, industry ties)
• Credibility scores based on funding independence (not credentials)
• Counter-perspectives from alternative sources
• Red flags indicating propaganda or biased reporting

Aletheia prioritizes funding transparency over credentials because a PhD funded by Pfizer is less trustworthy than an independent journalist citing peer-reviewed research. The tool is built for researchers, truth-seekers, and anyone refusing to be victimized by money-hungry entities pushing life-threatening narratives.

The Problem: Misinformation Crisis

The health information ecosystem is fundamentally broken:

1. Corporate Capture of Science

• Pharmaceutical companies fund medical schools and research
• Food industry sponsors nutrition studies with predetermined conclusions
• "Independent" experts receive consulting fees from corporations they defend
• Medical journals depend on pharma advertising revenue

2. Credential Washing

• A Harvard professor paid by Coca-Cola tells you sugar is fine
• An MD funded by supplement companies promotes unproven cures
• Government agencies staffed by former industry executives push industry agendas
• Credentials (PhD, MD) have become marketing tools, not truth indicators

3. Manufactured Consensus

• 100 news sites copy the same industry-funded press release
• Looks like "consensus" but it's just one narrative from six media conglomerates
• Counter-evidence is suppressed or labeled "conspiracy theory"
• Independent researchers struggle for funding and visibility

4. Life-Threatening Consequences

• Millions harmed by opioid crisis (pushed by "trusted" doctors)
• Dietary guidelines written by food industry lobbyists
• Suppression of generic drug alternatives for profit
• Medical interventions with hidden side effects

Existing fact-checkers often work for the same corporate interests. Traditional media literacy fails because it trusts credentials over funding. We need a tool that follows the money.

Core Goals

1. Expose Funding Conflicts

Reveal who pays for the research, who owns the publication, and what financial interests are at stake. Funding transparency is the #1 credibility factor.

2. Prioritize Independence Over Credentials
A PhD funded by industry is less credible than an independent researcher with proper sourcing. Credentials mean nothing without funding independence.

3. Find Counter-Perspectives
Break echo chambers by automatically finding alternative viewpoints from sources with different funding structures. Challenge single narratives.

4. Empower Critical Thinking
Provide users with the tools to question authority, follow the money, and make informed decisions about their health without being victimized by propaganda.

5. Build a Truth Database
Create a growing database of publication credibility scores based on funding analysis, available to all users and researchers.

Current Implementation (Phase 1 - MVP)

Architecture

Frontend: Chrome Extension
• Browser extension with sidebar interface
• User-triggered analysis (click extension icon to start)
• Displays real-time analysis results
• Tech: JavaScript, Chrome Extension API, HTML/CSS

Backend: Flask API Server
• RESTful API for article analysis
• Publication research and credibility scoring
• Database management (JSON for MVP, will migrate to PostgreSQL)
• Tech: Python, Flask, Flask-CORS

AI Research Layer
• Tavily API: Web search for publication information
• OpenAI GPT-4o-mini: Structured data extraction and analysis
• Smart caching: Avoids re-analyzing known publications

Data Storage
• publications.json: Growing database of analyzed sources
• myths.json: Common health myths database
• Auto-updates with each new analysis

Analysis Pipeline

When a user clicks the Aletheia extension icon on a health article, the system executes the following pipeline:

Step 1: Publication Research (1 Tavily API call)
• Extract domain from article URL
• Check cache for existing analysis
• If not cached or incomplete:
  - Search web for: ownership, funding, conflicts, editorial standards
  - Scrape publication's about page
  - Use AI to extract structured data
• Calculate credibility score (0-10) based on funding formula

Step 2: Myth Detection
• Scan article title against known health myths database

• Flag common misinformation patterns
• Examples: "superfood cures cancer", "detox cleanses", "toxins"

Step 3: Counter-Perspective Search (1-2 Tavily API calls)
• Search for alternative viewpoints with 3 strategies:
  - "[topic] fact check scientific evidence"
  - "[topic] medical research consensus"
  - "[topic] peer reviewed study"
• Filter candidates using promise score (avoid spam/commercial sites)
• Prioritize: medical journals > fact-checkers > research institutions
• Analyze best candidate (if credibility score >= 4)

Step 4: Comparative Analysis
• Compare main source vs counter-perspective:
  - Credibility gap calculation
  - Funding model comparison (independent vs corporate)
  - Red flags vs green flags
  - Conflicts of interest analysis
• Generate warnings and recommendations

Step 5: Display Results
• Show credibility scores with color coding (red/yellow/green)
• Display funding sources and conflicts
• Present counter-perspective with comparison
• Highlight critical missing context

Credibility Scoring Formula

Aletheia uses a funding-first approach to credibility (0-10 scale):

Start at 5.0 (neutral)

Primary Source Verification:
+ 2.0: Consistently links to peer-reviewed studies
+ 1.0: Sometimes links to sources
- 2.0: Rarely links to original research
- 3.0: Makes claims without any citations

Funding & Conflicts (Most Critical):
+ 2.0: Independent/nonprofit, no corporate ties
+ 1.5: Subscription-based, no advertising
- 1.0: Heavy advertising, limited disclosure
- 2.0: Pharmaceutical/supplement sponsorship

- 3.0: Owned by industry they cover
- 4.0: Hidden conflicts, undisclosed ties

Track Record:
+ 1.0: Zero retractions, strong fact-check history
- 2.0: History of retractions
- 3.0: Multiple fact-checker warnings

Transparency:
+ 1.0: Full disclosure of funding and conflicts
- 1.0: Vague or missing disclosures

Credentials (Minor Weight):
+ 1.0: Peer-reviewed medical journal
+ 0.5: MD/PhD AND independent funding
+ 0.0: MD/PhD BUT corporate-funded (credentials don't matter if bought)

Key Principle: Funding independence > Impressive credentials

Future Implementations

Phase 2: Enhanced Database & Topic Clustering (Q1 2026)
• Migrate from JSON to PostgreSQL for scalability
• Topic extraction and categorization ("cancer cure", "breakfast myths", etc.)
• Build pro/con databases for specific claims
• Track consensus across multiple sources per topic
• Show "15 sources debunk this, 2 support it" summaries

Phase 3: Community Features (Q2 2026)
• User submissions: Report suspicious publications
• Community voting on credibility assessments
• Researcher contributions to database
• Discussion forums for specific health topics
• Crowdsourced funding verification

Phase 4: Advanced AI Analysis (Q3 2026)
• Full article content analysis (not just publication)
• Claim extraction and verification
• Detect misleading statistics and cherry-picked data
• Identify logical fallacies in arguments
• Cross-reference claims with peer-reviewed research

Phase 5: Blockchain Integration (Q4 2026)

• Immutable credibility score history (Sui blockchain)
• Decentralized database (cannot be censored)
• NFT-based researcher credentials (verify independent status)
• Transparent funding trails on-chain
• Community governance via token voting

Phase 6: Multi-Platform Expansion (2027)
• Firefox and Edge extensions
• Mobile apps (iOS/Android)
• API for third-party integrations
• Embeddable widgets for websites
• Social media integration (analyze shared health links)

Phase 7: Real-Time Monitoring (2027)
• Track publication credibility changes over time
• Alert users when sources lose credibility
• Monitor funding changes (acquisition, new sponsors)
• Detect emerging misinformation campaigns
• Weekly credibility reports

Phase 8: Global Expansion (2027+)
• Multi-language support
• International publication databases
• Region-specific health myth detection
• Cross-cultural funding analysis
• Partnership with international truth-seeking organizations

Current Technical Specifications

API Usage & Costs
• Tavily API: 1,000 requests/month (free tier)
  - ~3 calls per article analysis
  - Supports ~333 article analyses/month
  - ~11 analyses per day
• OpenAI GPT-4o-mini: $5-10 budget
  - ~$0.0005 per analysis
  - $5 = ~10,000 analyses
  - Extremely cost-effective for MVP

Performance
• Average analysis time: 10-15 seconds
• Cache hit rate: Improves with usage (instant for cached sources)
• Database size: Growing with each analysis
• Concurrent users: Supports multiple simultaneous analyses

Data Structure (JSON Schema)

Publication Schema:


```json
{
  "domain": "medicalnewstoday.com",
  "name": "Medical News Today",
  "credibility_score": 6.5,
  "primary_source_links": true,
  "funding_transparency": "medium",
  "ownership": "Healthline Media UK Ltd.",
  "funding_sources": ["Advertising", "Affiliate links"],
  "conflicts_of_interest": [],
  "industry_ties": ["Health media conglomerate"],
  "author_credentials": "Medical review by MDs",
  "peer_reviewed": false,
  "editorial_standards": "Medical review process",
  "retraction_history": 0,
  "fact_check_rating": "Generally reliable",
  "red_flags": ["Heavy advertising"],
  "green_flags": ["Links to studies", "Medical review"],
  "expertise_areas": ["health", "medical", "nutrition"],
  "last_updated": "2025-12-28"
}
```


Example Analysis

Scenario: User visits article "Miracle Supplement Cures Cancer!"

Main Source Analysis:
Domain: naturalhealthcures.com
Credibility Score: 2.5/10 (LOW)
Funding: Supplement sales, affiliate links
Conflicts: Sells products mentioned in articles
Red Flags:
- No primary source links
- Owned by supplement company
- No medical credentials
- Clickbait headlines

Green Flags: None

Counter-Perspective Found:
Domain: cancer.gov (National Cancer Institute)
Credibility Score: 8.5/10 (HIGH)

Funding: Federal government, independent
Conflicts: None
Article: "No Evidence for Miracle Cancer Cures"
Green Flags:
• Links to peer-reviewed studies
• Government health agency
• No commercial interests
• Evidence-based content

Aletheia Warning:
"CRITICAL: 6.0 point credibility gap detected. Main source is commercial supplement site with undisclosed conflicts. Counter-source from independent government agency presents contradictory evidence. Recommendation: Trust the independent source with higher credibility."

User Outcome: Avoids buying dangerous "miracle cure", potentially saves life.

Why Aletheia Matters

The Mission
Aletheia exists to protect people from life-threatening misinformation spread by powerful entities prioritizing profit over truth. Whether it's pharmaceutical companies suppressing generic alternatives, food industry lobbyists writing dietary guidelines, or supplement companies selling snake oil—Aletheia follows the money to reveal the truth.

What Makes Aletheia Different

1. Funding > Credentials
Unlike traditional fact-checkers that trust credentials, Aletheia recognizes that PhDs and MDs can be bought. A Harvard professor funded by Coca-Cola is less credible than an independent journalist citing peer-reviewed research.

2. No Corporate Masters
Aletheia is built by independent developers, not funded by Facebook, Gates Foundation, or pharma companies. We have no conflicts of interest.

3. Counter-Narrative Focus
Rather than just saying "this is false", Aletheia actively finds and presents alternative perspectives from sources with different funding structures.

4. Transparent Methodology
All credibility scoring is based on verifiable factors: funding sources, conflicts of interest, primary source links. No black-box algorithms.

5. Growing Database

Every analysis contributes to a shared database of publication credibility, helping future users avoid propaganda.

Getting Started (Development Setup)

Prerequisites

- Python 3.8+
- Node.js (for future frontend development)
- Chrome browser
- API Keys: OpenAI, Tavily

Backend Setup

1. Clone repository:
```
git clone https://github.com/yourusername/aletheia.git
cd aletheia/backend
```

2. Create virtual environment:
```
python -m venv venv
source venv/bin/activate  # Windows: venv\Scripts\activate
```

3. Install dependencies:
```
pip install -r requirements.txt
```

4. Create .env file:
```
OPENAI_API_KEY=your_key_here
TAVILY_API_KEY=your_key_here
```

5. Run backend:
```
python app.py
# Server runs on http://localhost:5000
```

Extension Setup

1. Open Chrome and go to chrome://extensions/
2. Enable "Developer mode" (top right)
3. Click "Load unpacked"
4. Select the extension/ folder
5. Pin Aletheia icon to toolbar
6. Visit any health article to test

Project Status (December 2025)

Completed Features

✓ Chrome extension with sidebar UI
✓ Flask backend API
✓ Publication research and credibility scoring
✓ Funding transparency analysis
✓ Counter-perspective search
✓ Myth detection system
✓ Smart caching system
✓ JSON database (growing with each use)
✓ Comparative analysis (main vs counter-source)
✓ Red/green flag identification

In Progress
☐ UI/UX refinements
☐ Extended myth database
☐ Performance optimization
☐ Chrome Web Store preparation

Next Steps (Q1 2026)
☐ PostgreSQL migration
☐ User authentication
☐ Topic clustering
☐ Community features
☐ Public beta launch

Tech Stack Summary

Frontend:
• JavaScript (Vanilla)
• Chrome Extension API
• HTML5/CSS3

Backend:
• Python 3.10+
• Flask
• Flask-CORS
• BeautifulSoup4 (web scraping)

AI/APIs:
• OpenAI GPT-4o-mini
• Tavily Search API
• Feedparser (attempted Google News RSS)

Data:
• JSON (current)

• PostgreSQL (planned)
• Blockchain/Sui (future)

Key Learnings & Decisions

What We Learned
1. Funding matters more than credentials for credibility assessment
2. Google News RSS blocks automated redirect following (switched to Tavily)
3. Consensus can be manufactured - independent funding is better indicator
4. Users need ONE good counter-source, not five mediocre ones
5. Smart caching is essential for API cost management

Critical Design Decisions
1. Skip .gov/.edu blind trust - they can be captured by industry
2. Prioritize primary source links over author credentials
3. Use minimal filtering for counter-perspectives (let funding analysis decide)
4. Build growing database rather than real-time consensus
5. Focus on funding transparency as #1 feature

Challenges Overcome
• Google News RSS blocking → Switched to Tavily search
• NewsAPI limitations → Removed related articles feature for MVP
• Credential worship → Redesigned scoring to prioritize funding
• API quota concerns → Implemented smart caching and promise scoring
• Consensus complexity → Simplified to one quality counter-source

Conclusion

Aletheia represents a fundamental shift in how we evaluate health information. Rather than trusting authority figures and impressive credentials, Aletheia follows the money to reveal true motivations and conflicts of interest.

In a world where pharmaceutical companies fund medical education, food industry lobbyists write dietary guidelines, and supplement companies masquerade as health publishers, tools like Aletheia are not just useful—they're essential for survival.

Every day, people make life-or-death health decisions based on information from sources with undisclosed conflicts. Aletheia aims to change that, one article at a time.

The mission is simple: Expose the lies. Follow the money. Empower truth-seekers.

Contact & Contributions

Project: Aletheia ("Truth" in Greek)

Developer: Juan R
Institution: University of Alberta, Computing Science
GitHub: [Repository URL]
Email: [Your email]
Status: Active Development (Phase 1 MVP)
License: Open Source (TBD)

Contributions Welcome:
• Frontend developers
• Data scientists
• Medical researchers
• Fact-checkers
• Anyone passionate about exposing health misinformation

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